YUN-HO HAN

Ph.D. Candidate in Mechanical System Engineering

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RESEARCH INTERESTS

- Humanoid robot design
- Legged locomotion and bipedal walking
- Differential transmission mechanisms
- Whole-body control and model predictive control
- Reinforcement learning-based walking control
- Mechanism-aware reinforcement learning
- Sim-to-real and sim-to-sim validation
- Dynamic motion planning and kinodynamic trajectory optimization

TECHNICAL SKILLS

Robotics, Control, and Learning

- Humanoid robot design
- Legged locomotion control
- Whole-body control
- Model predictive control
- Torque-control-based walking
- Reinforcement learning-based control
- Trajectory optimization
- Nonlinear trajectory optimization
- Kinodynamic motion planning
- Sim-to-real validation
- Sim-to-sim validation

Simulation Tools

Isaac Gym | MuJoCo | Gazebo

Programming and Tools

Python | C/C++ | MATLAB | ROS / ROS 2 | Git / GitHub | Linux / Ubuntu | SolidWorks | LaTeX

RESEARCH PROFILE

I am a humanoid robotics researcher specializing in humanoid robot design, legged locomotion, differential transmission mechanisms, model-based walking control, and reinforcement learning-based bipedal walking control.

My research has progressed from RoK-3 humanoid walking experiments to RoK-4, a humanoid robot platform equipped with a Hip Pitch-Knee differential mechanism. I have worked on humanoid robot platforms, WBC/MPC-based walking control, dynamic motion planning, and mechanism-aware reinforcement learning for real-world humanoid locomotion.

My recent work focuses on the ADAPT Hip Pitch-Knee differential mechanism and mechanism-aware reinforcement learning-based walking control, validated through real-world humanoid walking experiments and full-mechanism simulation.

EDUCATION

Ph.D. Candidate in Mechanical System Engineering
Kookmin University, Seoul, Republic of Korea | Expected Graduation: August 2026
Dissertation: *Humanoid Robot Leg Design Based on a Hip Pitch-Knee Differential Mechanism and Reinforcement Learning-Based Walking Control Using the Designed Leg*

M.S. in Mechanical System Engineering
Kookmin University, Seoul, Republic of Korea | February 2020

B.S. in Mechanical System Engineering
Kookmin University, Seoul, Republic of Korea | February 2018

RESEARCH EXPERIENCE

Graduate Researcher, Humanoid Robotics and Legged Locomotion
Kookmin University, Seoul, Republic of Korea | 2018 - Present

- Conducted research on humanoid robot design, bipedal walking control, and legged locomotion.
- Worked on RoK-3 and RoK-4 humanoid platforms, covering walking experiments, mechanism design, model-based control, and learning-based control.
- Developed RoK-4, a humanoid robot platform equipped with a Hip Pitch-Knee differential mechanism.
- Designed and analyzed the ADAPT differential transmission mechanism for high-performance humanoid robot legs.
- Formalized actuator-joint position, velocity, and torque mapping for mechanism-aware control.
- Developed PPO-based reinforcement learning walking controllers using mechanism-aware observations, actions, and actuator-joint mapping.
- Validated learned walking policies through real-world humanoid walking experiments and MuJoCo full-mechanism simulation.

SELECTED RESEARCH PROJECTS

RoK-4 Humanoid Robot

Developed a humanoid robot platform for high-performance bipedal locomotion using a Hip Pitch-Knee differential leg mechanism.

- Participated in the design and development of the RoK-4 humanoid robot platform.
- Integrated a Hip Pitch-Knee differential mechanism into the robot leg structure.
- Developed a mechanism-aware reinforcement learning environment for bipedal walking.
- Demonstrated real-world walking and MuJoCo full-mechanism sim-to-sim validation.

Keywords: Humanoid robot, bipedal locomotion, differential mechanism, sim-to-real, sim-to-sim

ADAPT: Hip Pitch-Knee Differential Mechanism

Developed ADAPT, an Advanced Differential and Agile Parallel Transmission mechanism for humanoid robot legs.

- Designed a differential transmission structure coupling actuator inputs with Hip Pitch and Knee joint outputs.
- Formalized actuator-joint position, velocity, and torque mapping.
- Analyzed the torque-velocity capability of the mechanism.
- Implemented the mechanism in the RoK-4 humanoid robot leg.
- Validated the mechanism through humanoid walking experiments and full-mechanism simulation.

Keywords: Differential transmission, humanoid robot leg, actuator-joint mapping, torque-velocity capability

Mechanism-Aware Reinforcement Learning for Humanoid Walking

Developed a reinforcement learning-based walking controller that explicitly considers the actuator-joint relationship of the ADAPT mechanism.

- Built a mechanism-aware reinforcement learning environment in Isaac Gym.
- Designed observation and action spaces using actuator-space and joint-space information.
- Applied actuator-joint mapping for action conversion and torque transformation.
- Trained walking policies using PPO with an asymmetric actor-critic architecture.
- Validated learned policies through real-world experiments and MuJoCo full-mechanism simulation.

Keywords: Reinforcement learning, PPO, humanoid walking, asymmetric actor-critic, actuator-joint mapping

Model-Based Humanoid Walking Control

Studied model-based walking control approaches for humanoid robots, including WBC, MPC, and torque-control-based walking.

- Studied whole-body control frameworks for humanoid balancing and locomotion.
- Applied model predictive control concepts for bipedal walking and stability.
- Investigated torque-control-based walking with high-reduction gears and no joint torque feedback.
- Built a foundation for later mechanism-aware reinforcement learning research.

Keywords: Whole-body control, model predictive control, torque control, humanoid walking

RoK-3 Humanoid Robot

Worked on RoK-3, an earlier humanoid platform for bipedal walking experiments and locomotion research.

- Conducted humanoid walking experiments using the RoK-3 platform.
- Studied walking stability, foot mechanism design, terrain adaptation, variable stiffness soles, and slope walking.

Keywords: RoK-3, humanoid walking, foot mechanism, slope walking

Single-Leg Jumping Motion Planning

Developed dynamic motion planning methods for agile legged robot motion using kinodynamic trajectory optimization.

- Formulated jumping motion generation as a nonlinear trajectory optimization problem.
- Generated and validated dynamically feasible jumping trajectories for a single-leg platform.

Keywords: Nonlinear trajectory optimization, kinodynamic motion planning, jumping motion

YUN-HO HAN

Publications, Awards, and Demonstrations

PUBLICATIONS

International Journal Papers

1. **Yun-Ho Han**, Min-Ho Park, and Baek-Kyu Cho, "ADAPT: Advanced Differential and Agile Parallel Transmission Mechanism for High-Performance Humanoid Robots," *IEEE Robotics and Automation Letters*, 2026, pp. 1-8. doi: 10.1109/LRA.2026.3686707.
2. Junyeon Namgung, **Yun-Ho Han**, and Baek-Kyu Cho, "A Variable Stiffness Sole for Biped Robot and Its Experimental Verification," *ASME Journal of Mechanisms and Robotics*, vol. 16, no. 5, Article 054502, 2024. doi: 10.1115/1.4062650.
3. **Yun-Ho Han** and Baek-Kyu Cho, "Slope Walking of Humanoid Robot without IMU Sensor on an Unknown Slope," *Robotics and Autonomous Systems*, vol. 155, Article 104163, 2022. doi: 10.1016/j.robot.2022.104163.
4. **Yun-Ho Han**, Ho-Jin Jeon, and Baek-Kyu Cho, "Development of a Humanoid Robot for the 2018 Ski Robot Challenge," *International Journal of Precision Engineering and Manufacturing*, vol. 21, no. 7, pp. 1309-1320, 2020. doi: 10.1007/s12541-020-00344-6.

International Conference Papers

1. **Yun-Ho Han** and Baek-Kyu Cho, "Implementing Torque Control-Based Biped Walking of Humanoid Robots with High Reduction Gear and No Joint Torque Feedback," *2023 IEEE-RAS 22nd International Conference on Humanoid Robots (Humanoids)*, Austin, TX, USA, 2023, pp. 1-8. doi: 10.1109/Humanoids57100.2023.10375177.
2. **Yun-Ho Han**, Junyeon Namgung, and Baek-Kyu Cho, "Walking Control Framework on Uneven Terrain Using Variable Stiffness Sole," *2022 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, Kyoto, Japan, 2022, pp. 8328-8335.

Domestic Conference / Workshop Papers

1. **Yun-Ho Han**, So-Yeon Oh, and Baek-Kyu Cho, "Humanoid Robot Legs Designed with Four-Bar Link Structure," *12th Korea Robotics Society Annual Conference (KRoC)*, Pyeongchang, Republic of Korea, 2017.
2. **Yun-Ho Han** and Baek-Kyu Cho, "Walking Humanoid Robot Controlled by Complementary Filter and PD Control," *11th Korea Robotics Society Annual Conference (KRoC)*, Pyeongchang, Republic of Korea, 2016.

AWARDS AND HONORS

1. **Best Paper Award**, Dynamics and Control / Robotics Division, Spring Conference of the Korean Society of Mechanical Engineers, 2025. Yun-Ho Han and Baek-Kyu Cho.
2. **Grand Prize**, RED Show, 20th Korea Robotics Society Annual Conference (KRoC 2025), 2025. Yun-Ho Han, Min-Ho Park, and Baek-Kyu Cho.
3. **Best Paper Award**, IT Convergence Division, Spring Conference of the Korean Society of Mechanical Engineers, 2024. Yun-Ho Han, Min-Ho Park, and Baek-Kyu Cho.

SELECTED DEMONSTRATIONS

- RoK-3 humanoid walking
- RoK-4 real-world walking
- RoK-4 full-mechanism MuJoCo simulation
- ADAPT differential mechanism demonstration
- Reinforcement learning-based humanoid walking
- Single-leg jumping motion generation

REFERENCES

Available upon request.